



DHA SUFFA UNIVERSITY

Off Khayaban-e-Tufail, Phase -VII (Extension), DHA, Karachi - 75500

Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

BS-SE-2020

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1103	Applied Physics	2
2	BS-1301	Calculus and Analytical Geometry	3
3	HU-1002	English Composition and Comprehension	3
4	HU-2101	Pakistan Studies	2
5	SE-1001	Programming Fundamentals	3
6	SE-1001L	Programming Fundamentals Lab	1
7	SE-1201	Introduction to Information and Communication Technology	3
8	SE-1201L	Introduction to Information and Communication Technology Lab	1
Total			18

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	SE-2001	Data Structures & Algorithms	3
3	SE-2001L	Data Structures & Algorithms Lab	1
4	SE-2401	Software Requirement Engineering	3
5	SE-2402	Human Computer Interaction	3
6	UE-2XXX	University Elective II	3
Total			16

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	HU-2009	Technical and Business Writing	3
2	SE-3001	Computer Networks	3
3	SE-3001L	Computer Networks Lab	1
4	SE-3401	Software Construction & Development	2
5	SE-3401L	Software Construction & Development Lab	1
6	SE-3501	Business Process Engineering	3
7	SE-3502	Simulation & Modeling	3
Total			16

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-2201	Professional Practices	3
2	SE-4001	Final Year Project - I	3
3	SE-4401	Software Project Management	3
4	SE-XXXX	SE Elective II	3
5	SE-XXXX	SE Elective III	3
Total			15

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	HU-2001	Communication and Presentation Skills	3
2	HU-2201	Islamic Studies	2
3	SE-1002	Object Oriented Programming	3
4	SE-1002L	Object Oriented Programming Lab	1
5	SE-1003	Discrete Structures	3
6	SE-1004	Software Engineering	3
7	UE-1XXX	University Elective I	3
Total			18

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	SE-2003	Operating Systems	3
3	SE-2003L	Operating Systems Lab	1
4	SE-2004	Database Systems	3
5	SE-2004L	Database Systems Lab	1
6	SE-2403	Software Design & Architecture	2
7	SE-2403L	Software Design & Architecture Lab	1
8	UE-XXXX	University Elective III	3
Total			17

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	SE-3002	Information Security	3
2	SE-3402	Software Quality Engineering	3
3	SE-3403	Web Engineering	3
4	SE-3503	Operation Research	3
5	SE-XXXX	SE Elective I	3
Total			15

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	SE-4002	Final Year Project - II	3
2	SE-4402	Software Re-Engineering	3
3	SE-4XXX	SE Elective IV	3
4	SE-4XXX	SE Elective V	3
5	UE-XXXX	University Elective IV	3
Total			15

Total Credit Hours Required: 130



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Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

BS-SE-2020 (Minorities)

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1103	Applied Physics	2
2	BS-1301	Calculus and Analytical Geometry	3
3	HU-1002	English Composition and Comprehension	3
4	HU-2101	Pakistan Studies	2
5	SE-1001	Programming Fundamentals	3
6	SE-1001L	Programming Fundamentals Lab	1
7	SE-1201	Introduction to Information and Communication Technology	3
8	SE-1201L	Introduction to Information and Communication Technology Lab	1
Total			18

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	SE-2001	Data Structures & Algorithms	3
3	SE-2001L	Data Structures & Algorithms Lab	1
4	SE-2401	Software Requirement Engineering	3
5	SE-2402	Human Computer Interaction	3
6	SE-XXXX	University Elective II	3
Total			16

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	HU-2009	Technical and Business Writing	3
2	SE-3001	Computer Networks	3
3	SE-3001L	Computer Networks Lab	1
4	SE-3401	Software Construction & Development	2
5	SE-3401L	Software Construction & Development Lab	1
6	SE-3501	Business Process Engineering	3
7	SE-3502	Simulation & Modeling	3
Total			16

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-2201	Professional Practices	3
2	SE-4001	Final Year Project - I	3
3	SE-4401	Software Project Management	3
4	SE-XXXX	SE Elective II	3
5	SE-XXXX	SE Elective III	3
Total			15

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	HU-1603	Ethics	2
2	HU-2001	Communication and Presentation Skills	3
3	SE-1002	Object Oriented Programming	3
4	SE-1002L	Object Oriented Programming Lab	1
5	SE-1003	Discrete Structures	3
6	SE-1004	Software Engineering	3
7	UE-1XXX	University Elective I	3
Total			18

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	SE-2003	Operating Systems	3
3	SE-2003L	Operating Systems Lab	1
4	SE-2004	Database Systems	3
5	SE-2004L	Database Systems Lab	1
6	SE-2403	Software Design & Architecture	2
7	SE-2403L	Software Design & Architecture Lab	1
8	SE-XXXX	University Elective III	3
Total			17

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	SE-3002	Information Security	3
2	SE-3402	Software Quality Engineering	3
3	SE-3403	Web Engineering	3
4	SE-3503	Operation Research	3
5	SE-XXXX	SE Elective I	3
Total			15

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	SE-4002	Final Year Project - II	3
2	SE-4402	Software Re-Engineering	3
3	SE-4XXX	SE Elective IV	3
4	SE-4XXX	SE Elective V	3
5	SE-XXXX	University Elective IV	3
Total			15

Total Credit Hours Required: 130



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Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

BS-SE-2020 (Minorities)

PreMed

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1002	Elementary Mathematics I	0
2	BS-1103	Applied Physics	2
3	BS-1301	Calculus and Analytical Geometry	3
4	HU-1002	English Composition and Comprehension	3
5	HU-2101	Pakistan Studies	2
6	SE-1001	Programming Fundamentals	3
7	SE-1001L	Programming Fundamentals Lab	1
8	SE-1201	Introduction to Information and Communication Technology	3
9	SE-1201L	Introduction to Information and Communication Technology Lab	1
Total			18

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	SE-2001	Data Structures & Algorithms	3
3	SE-2001L	Data Structures & Algorithms Lab	1
4	SE-2401	Software Requirement Engineering	3
5	SE-2402	Human Computer Interaction	3
6	SE-XXXX	University Elective II	3
Total			16

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	HU-2009	Technical and Business Writing	3
2	SE-3001	Computer Networks	3
3	SE-3001L	Computer Networks Lab	1
4	SE-3401	Software Construction & Development	2
5	SE-3401L	Software Construction & Development Lab	1
6	SE-3501	Business Process Engineering	3
7	SE-3502	Simulation & Modeling	3
Total			16

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-2201	Professional Practices	3
2	SE-4001	Final Year Project - I	3
3	SE-4401	Software Project Management	3
4	SE-XXXX	SE Elective II	3
5	SE-XXXX	SE Elective III	3
Total			15

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1003	Elementary Mathematics II	0
2	HU-1603	Ethics	2
3	HU-2001	Communication and Presentation Skills	3
4	SE-1002	Object Oriented Programming	3
5	SE-1002L	Object Oriented Programming Lab	1
6	SE-1003	Discrete Structures	3
7	SE-1004	Software Engineering	3
8	UE-1XXX	University Elective I	3
Total			18

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	SE-2003	Operating Systems	3
3	SE-2003L	Operating Systems Lab	1
4	SE-2004	Database Systems	3
5	SE-2004L	Database Systems Lab	1
6	SE-2403	Software Design & Architecture	2
7	SE-2403L	Software Design & Architecture Lab	1
8	SE-XXXX	University Elective III	3
Total			17

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	SE-3002	Information Security	3
2	SE-3402	Software Quality Engineering	3
3	SE-3403	Web Engineering	3
4	SE-3503	Operation Research	3
5	SE-XXXX	SE Elective I	3
Total			15

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	SE-4002	Final Year Project - II	3
2	SE-4402	Software Re-Engineering	3
3	SE-4XXX	SE Elective IV	3
4	SE-4XXX	SE Elective V	3
5	SE-XXXX	University Elective IV	3
Total			15

Total Credit Hours Required: 130



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Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

BS-SE-2020 - Special

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1103	Applied Physics	2
2	BS-1301	Calculus and Analytical Geometry	3
3	HU-1002	English Composition and Comprehension	3
4	HU-2101	Pakistan Studies	2
5	SE-1001	Programming Fundamentals	3
6	SE-1001L	Programming Fundamentals Lab	1
7	SE-1201	Introduction to Information and Communication Technology	3
8	SE-1201L	Introduction to Information and Communication Technology Lab	1
Total			18

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	SE-2001	Data Structures & Algorithms	3
3	SE-2001L	Data Structures & Algorithms Lab	1
4	SE-2401	Software Requirement Engineering	3
5	SE-2402	Human Computer Interaction	3
6	UE-2XXX	University Elective II	3
Total			16

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	HU-2009	Technical and Business Writing	3
2	SE-3001	Computer Networks	3
3	SE-3001L	Computer Networks Lab	1
4	SE-3401	Software Construction & Development	2
5	SE-3401L	Software Construction & Development Lab	1
6	SE-3501	Business Process Engineering	3
7	SE-3502	Simulation & Modeling	3
Total			16

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-2201	Professional Practices	3
2	SE-4001	Final Year Project - I	3
3	SE-4401	Software Project Management	3
4	SE-XXXX	SE Elective II	3
5	SE-XXXX	SE Elective III	3
Total			15

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	HU-2001	Communication and Presentation Skills	3
2	HU-2201	Islamic Studies	2
3	SE-1002	Object Oriented Programming	3
4	SE-1002L	Object Oriented Programming Lab	1
5	SE-1003	Discrete Structures	3
6	SE-1004	Software Engineering	3
7	UE-1XXX	University Elective I	3
Total			18

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	SE-2003	Operating Systems	3
3	SE-2003L	Operating Systems Lab	1
4	SE-2004	Database Systems	3
5	SE-2004L	Database Systems Lab	1
6	SE-2403	Software Design & Architecture	2
7	SE-2403L	Software Design & Architecture Lab	1
8	UE-XXXX	University Elective III	3
Total			17

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	SE-3002	Information Security	3
2	SE-3402	Software Quality Engineering	3
3	SE-3403	Web Engineering	3
4	SE-3503	Operation Research	3
5	SE-XXXX	SE Elective I	3
Total			15

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	SE-4002	Final Year Project - II	3
2	SE-4402	Software Re-Engineering	3
3	SE-4XXX	SE Elective IV	3
4	SE-4XXX	SE Elective V	3
5	UE-XXXX	University Elective IV	3
Total			15

Total Credit Hours Required: 130



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Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

BS-SE-2020 PreMed

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1002	Elementary Mathematics I	0
2	BS-1103	Applied Physics	2
3	BS-1301	Calculus and Analytical Geometry	3
4	HU-1002	English Composition and Comprehension	3
5	HU-2101	Pakistan Studies	2
6	SE-1001	Programming Fundamentals	3
7	SE-1001L	Programming Fundamentals Lab	1
8	SE-1201	Introduction to Information and Communication Technology	3
9	SE-1201L	Introduction to Information and Communication Technology Lab	1
Total			18

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	SE-2001	Data Structures & Algorithms	3
3	SE-2001L	Data Structures & Algorithms Lab	1
4	SE-2401	Software Requirement Engineering	3
5	SE-2402	Human Computer Interaction	3
6	UE-2XXX	University Elective II	3
Total			16

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	HU-2009	Technical and Business Writing	3
2	SE-3001	Computer Networks	3
3	SE-3001L	Computer Networks Lab	1
4	SE-3401	Software Construction & Development	2
5	SE-3401L	Software Construction & Development Lab	1
6	SE-3501	Business Process Engineering	3
7	SE-3502	Simulation & Modeling	3
Total			16

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-2201	Professional Practices	3
2	SE-4001	Final Year Project - I	3
3	SE-4401	Software Project Management	3
4	SE-XXXX	SE Elective II	3
5	SE-XXXX	SE Elective III	3
Total			15

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1003	Elementary Mathematics II	0
2	HU-2001	Communication and Presentation Skills	3
3	HU-2201	Islamic Studies	2
4	SE-1002	Object Oriented Programming	3
5	SE-1002L	Object Oriented Programming Lab	1
6	SE-1003	Discrete Structures	3
7	SE-1004	Software Engineering	3
8	UE-1XXX	University Elective I	3
Total			18

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	SE-2003	Operating Systems	3
3	SE-2003L	Operating Systems Lab	1
4	SE-2004	Database Systems	3
5	SE-2004L	Database Systems Lab	1
6	SE-2403	Software Design & Architecture	2
7	SE-2403L	Software Design & Architecture Lab	1
8	UE-XXXX	University Elective III	3
Total			17

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	SE-3002	Information Security	3
2	SE-3402	Software Quality Engineering	3
3	SE-3403	Web Engineering	3
4	SE-3503	Operation Research	3
5	SE-XXXX	SE Elective I	3
Total			15

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	SE-4002	Final Year Project - II	3
2	SE-4402	Software Re-Engineering	3
3	SE-4XXX	SE Elective IV	3
4	SE-4XXX	SE Elective V	3
5	UE-XXXX	University Elective IV	3
Total			15

Total Credit Hours Required: 130



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Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

BS-SE-2024

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1101	Applied Physics	3
2	BS-1301	Calculus and Analytical Geometry	3
3	CS-1001	Applications of Information and Communication Technologies	2
4	CS-1001L	Applications of Information and Communication Technologies L	1
5	CS-1002	Programming Fundamentals	3
6	CS-1002L	Programming Fundamentals Lab	1
7	HU-1002	Functional English	3
8	HU-1203	Islamic Studies	2
Total			18

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	CS-1003	Discrete Structures	3
3	CS-1004	Digital Logic Design	2
4	CS-1004L	Digital Logic Design Lab	1
5	CS-1005	Object Oriented Programming	3
6	CS-1005L	Object Oriented Programming Lab	1
7	HU-2101	Pakistan Studies	2
8	HU-2203	Expository Writing	3
Total			18

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-2301	Multivariate Calculus	3
2	CS-2007	Data Structures & Algorithms	3
3	CS-2007L	Data Structures & Algorithms Lab	1
4	CS-2201	Computer Networks	3
5	CS-2201L	Computer Networks Lab	1
6	CS-2801	Software Engineering	3
7	MG-2XXX	University Elective-I (OB)	2
Total			16

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	CS-2008	Computer Organization & Assembly Language	2
3	CS-2008L	Computer Organization & Assembly Language Lab	1
4	CS-2009	Operating Systems	3
5	CS-2009L	Operating Systems Lab	1
6	CS-2101	Database Systems	3
7	CS-2101L	Database Systems Lab	1
8	CS-2202	Information Security	2
9	CS-2202L	Information Security Lab	1
Total			17

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	CS-3301	Artificial Intelligence	2
2	CS-3301L	Artificial Intelligence Lab	1
3	SE-3801	Software Requirement Engineering	3
4	SE-3802	Software Quality Engineering	2
5	SE-3802L	Software Quality Engineering Lab	1
6	SE-XXXX	SE Elective I	3
7	SE-XXXX	SE Elective II	3
Total			15

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	CS-3011	Analysis of Algorithms	3
2	HU-2104	Ideology and Constitution of Pakistan	2
3	HU-31XX	Fehm-e-Quran – I	1
4	SE-3601	Parallel and Distributed Computing	3
5	SE-3803	Software Design & Architecture	3
6	SE-3804	Software Project Management	3
7	SE-XXXX	SE Elective III	3
Total			18

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-4013	Final Year Project-I	3
2	HU-2009	Technical and Business Writing	3
3	HU-32XX	Fehm-e-Quran – II	1
4	SE-4806	Software Construction and Development	2
5	SE-4806L	Software Construction and Development Lab	1
6	SE-4XXX	SE Elective V	3
7	SE-4XXX	SE Elective IV	3
Total			16

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	CS-4014	Final Year Project - II	3
2	HU-2200	Civics and Community Engagement	2
3	HU-3041	Professional Practice	2
4	HU-XXXX	University Elective II	3
5	MS-4111	Entrepreneurship	3
6	SE-4XXX	SE Elective VII	3
7	SE-4XXX	SE Elective VI	3
Total			19

Total Credit Hours Required: 137



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Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

BS-SE-2024 (Minorities)

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1101	Applied Physics	3
2	BS-1301	Calculus and Analytical Geometry	3
3	CS-1001	Applications of Information and Communication Technologies	2
4	CS-1001L	Applications of Information and Communication Technologies L	1
5	CS-1002	Programming Fundamentals	3
6	CS-1002L	Programming Fundamentals Lab	1
7	HU-1002	Functional English	3
8	HU-1603	Ethics	2
Total			18

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	CS-1003	Discrete Structures	3
3	CS-1004	Digital Logic Design	2
4	CS-1004L	Digital Logic Design Lab	1
5	CS-1005	Object Oriented Programming	3
6	CS-1005L	Object Oriented Programming Lab	1
7	HU-2101	Pakistan Studies	2
8	HU-2203	Expository Writing	3
Total			18

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-2301	Multivariate Calculus	3
2	CS-2007	Data Structures & Algorithms	3
3	CS-2007L	Data Structures & Algorithms Lab	1
4	CS-2201	Computer Networks	3
5	CS-2201L	Computer Networks Lab	1
6	CS-2801	Software Engineering	3
7	MG-2XXX	University Elective-I (OB)	2
Total			16

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	CS-2008	Computer Organization & Assembly Language	2
3	CS-2008L	Computer Organization & Assembly Language Lab	1
4	CS-2009	Operating Systems	3
5	CS-2009L	Operating Systems Lab	1
6	CS-2101	Database Systems	3
7	CS-2101L	Database Systems Lab	1
8	CS-2202	Information Security	2
9	CS-2202L	Information Security Lab	1
Total			17

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	CS-3301	Artificial Intelligence	2
2	CS-3301L	Artificial Intelligence Lab	1
3	SE-3801	Software Requirement Engineering	3
4	SE-3802	Software Quality Engineering	2
5	SE-3802L	Software Quality Engineering Lab	1
6	SE-XXXX	SE Elective I	3
7	SE-XXXX	SE Elective II	3
Total			15

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	CS-3011	Analysis of Algorithms	3
2	HU-1425	Religion, Culture and Society	2
3	HU-2104	Ideology and Constitution of Pakistan	2
4	SE-3601	Parallel and Distributed Computing	3
5	SE-3803	Software Design & Architecture	3
6	SE-3804	Software Project Management	3
7	SE-XXXX	SE Elective III	3
Total			19

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-4013	Final Year Project-I	3
2	HU-2009	Technical and Business Writing	3
3	SE-4806	Software Construction and Development	2
4	SE-4806L	Software Construction and Development Lab	1
5	SE-XXXX	SE Elective V	3
6	SE-XXXX	SE Elective IV	3
Total			15

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	CS-4014	Final Year Project - II	3
2	HU-2200	Civics and Community Engagement	2
3	HU-3041	Professional Practice	2
4	HU-XXXX	University Elective II	3
5	MS-4111	Entrepreneurship	3
6	SE-4XXX	SE Elective VII	3
7	SE-4XXX	SE Elective VI	3
Total			19

Total Credit Hours Required: 137



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Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

BS-SE-2024 (Minorities)

PreMed

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1002	Elementary Mathematics I	0
2	BS-1101	Applied Physics	3
3	CS-1001	Applications of Information and Communication Technologies	2
4	CS-1001L	Applications of Information and Communication Technologies L	1
5	CS-1002	Programming Fundamentals	3
6	CS-1002L	Programming Fundamentals Lab	1
7	HU-1002	Functional English	3
8	HU-1603	Ethics	2
Total			15

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	CS-2007	Data Structures & Algorithms	3
3	CS-2007L	Data Structures & Algorithms Lab	1
4	CS-2201	Computer Networks	3
5	CS-2201L	Computer Networks Lab	1
6	CS-2801	Software Engineering	3
7	MG-2XXX	University Elective-I (OB)	2
Total			16

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	CS-3301	Artificial Intelligence	2
3	CS-3301L	Artificial Intelligence Lab	1
4	SE-3801	Software Requirement Engineering	3
5	SE-3802	Software Quality Engineering	2
6	SE-3802L	Software Quality Engineering Lab	1
7	SE-XXXX	SE Elective II	3
8	SE-XXXX	SE Elective I	3
Total			18

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-4013	Final Year Project-I	3
2	HU-2009	Technical and Business Writing	3
3	SE-4806	Software Construction and Development	2
4	SE-4806L	Software Construction and Development Lab	1
5	SE-XXXX	SE Elective V	3
6	SE-XXXX	SE Elective IV	3
Total			15

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1003	Elementary Mathematics II	0
2	BS-1301	Calculus and Analytical Geometry	3
3	CS-1003	Discrete Structures	3
4	CS-1004	Digital Logic Design	2
5	CS-1004L	Digital Logic Design Lab	1
6	CS-1005	Object Oriented Programming	3
7	CS-1005L	Object Oriented Programming Lab	1
8	HU-2101	Pakistan Studies	2
9	HU-2203	Expository Writing	3
Total			18

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-2301	Multivariate Calculus	3
2	CS-2008	Computer Organization & Assembly Language	2
3	CS-2008L	Computer Organization & Assembly Language Lab	1
4	CS-2009	Operating Systems	3
5	CS-2009L	Operating Systems Lab	1
6	CS-2101	Database Systems	3
7	CS-2101L	Database Systems Lab	1
8	CS-2202	Information Security	2
9	CS-2202L	Information Security Lab	1
Total			17

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	CS-3011	Analysis of Algorithms	3
2	HU-1425	Religion, Culture and Society	2
3	HU-2104	Ideology and Constitution of Pakistan	2
4	SE-3601	Parallel and Distributed Computing	3
5	SE-3803	Software Design & Architecture	3
6	SE-3804	Software Project Management	3
7	SE-XXXX	SE Elective III	3
Total			19

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	CS-4014	Final Year Project - II	3
2	HU-2200	Civics and Community Engagement	2
3	HU-3041	Professional Practice	2
4	HU-XXXX	University Elective II	3
5	MS-4111	Entrepreneurship	3
6	SE-4XXX	SE Elective VII	3
7	SE-4XXX	SE Elective VI	3
Total			19

Total Credit Hours Required: 137



DHA SUFFA UNIVERSITY

Off Khayaban-e-Tufail, Phase -VII (Extension), DHA, Karachi - 75500

Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

BS-SE-2024 PreMed

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1002	Elementary Mathematics I	0
2	BS-1101	Applied Physics	3
3	CS-1001	Applications of Information and Communication Technologies	2
4	CS-1001L	Applications of Information and Communication Technologies L	1
5	CS-1002	Programming Fundamentals	3
6	CS-1002L	Programming Fundamentals Lab	1
7	HU-1002	Functional English	3
8	HU-1203	Islamic Studies	2
Total			15

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	CS-2007	Data Structures & Algorithms	3
3	CS-2007L	Data Structures & Algorithms Lab	1
4	CS-2201	Computer Networks	3
5	CS-2201L	Computer Networks Lab	1
6	CS-2801	Software Engineering	3
Total			14

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	CS-3301	Artificial Intelligence	2
3	CS-3301L	Artificial Intelligence Lab	1
4	SE-3801	Software Requirement Engineering	3
5	SE-3802	Software Quality Engineering	2
6	SE-3802L	Software Quality Engineering Lab	1
7	SE-XXXX	SE Elective I	3
8	SE-XXXX	SE Elective II	3
Total			18

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-4013	Final Year Project-I	3
2	HU-2009	Technical and Business Writing	3
3	HU-32XX	Fehm-e-Quran – II	1
4	MG-2XXX	University Elective-I (OB)	2
5	SE-4806	Software Construction and Development	2
6	SE-4806L	Software Construction and Development Lab	1
7	SE-4XXX	SE Elective V	3
8	SE-4XXX	SE Elective IV	3
Total			18

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1003	Elementary Mathematics II	0
2	BS-1301	Calculus and Analytical Geometry	3
3	CS-1003	Discrete Structures	3
4	CS-1004	Digital Logic Design	2
5	CS-1004L	Digital Logic Design Lab	1
6	CS-1005	Object Oriented Programming	3
7	CS-1005L	Object Oriented Programming Lab	1
8	HU-2101	Pakistan Studies	2
9	HU-2203	Expository Writing	3
Total			18

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-2301	Multivariate Calculus	3
2	CS-2008	Computer Organization & Assembly Language	2
3	CS-2008L	Computer Organization & Assembly Language Lab	1
4	CS-2009	Operating Systems	3
5	CS-2009L	Operating Systems Lab	1
6	CS-2101	Database Systems	3
7	CS-2101L	Database Systems Lab	1
8	CS-2202	Information Security	2
9	CS-2202L	Information Security Lab	1
Total			17

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	CS-3011	Analysis of Algorithms	3
2	HU-2104	Ideology and Constitution of Pakistan	2
3	HU-31XX	Fehm-e-Quran – I	1
4	SE-3601	Parallel and Distributed Computing	3
5	SE-3803	Software Design & Architecture	3
6	SE-3804	Software Project Management	3
7	SE-XXXX	SE Elective III	3
Total			18

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	CS-4014	Final Year Project - II	3
2	HU-2200	Civics and Community Engagement	2
3	HU-3041	Professional Practice	2
4	HU-XXXX	University Elective II	3
5	MS-4111	Entrepreneurship	3
6	SE-4XXX	SE Elective VII	3
7	SE-4XXX	SE Elective VI	3
Total			19

Total Credit Hours Required: 137